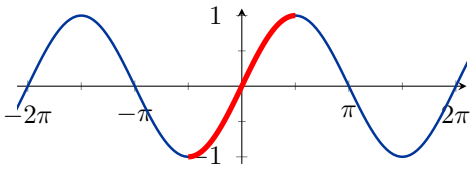
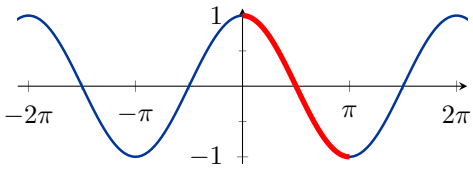
## 6.3: Inverse Trigonometric Functions

Note that $\sin(x)$ and $\cos(x)$,

- take in the measure of an angle, and
- return the ratio of the sides.

Recall that trigonometric functions are periodic:




Consider the following restricted domains and associated ranges:

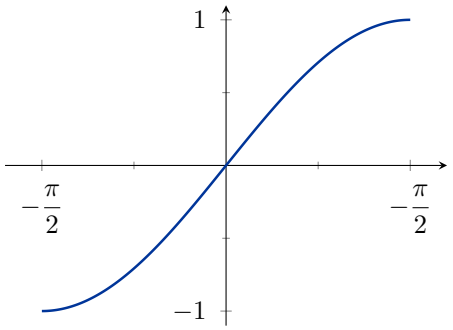
| | Domain | Range |
|---|---|---|
| $y = \sin(x)$ | $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ | $[-1, 1]$ |
| $y = \cos(x)$ | $[0, \pi]$ | $[-1, 1]$ |
| $y = \tan(x)$ | $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ | $(-\infty, \infty)$ |

When we use the restricted domains, then these functions are one-to-one. The resulting inverse functions are:

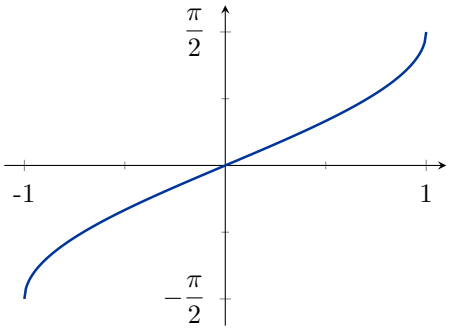
$$\arcsin(y) = \sin^{-1}(y)$$
$$\arccos(y) = \cos^{-1}(y)$$
$$\arctan(y) = \tan^{-1}(y)$$

**Example.** Since $\sin(0.96593) \approx \frac{5\pi}{12}$, what is $\sin^{-1}\left(\frac{4}{15}\right)$?

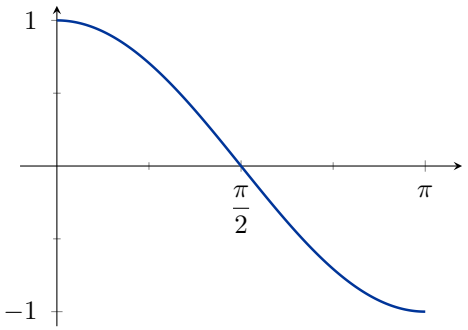
$y = \sin(x)$



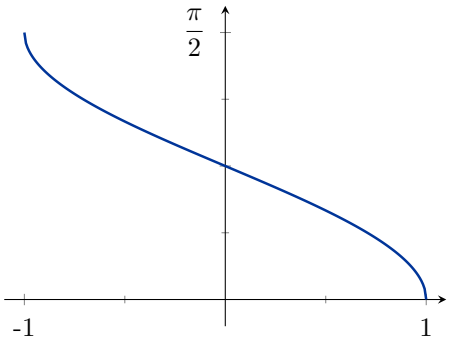

$y = \sin^{-1}(x)$



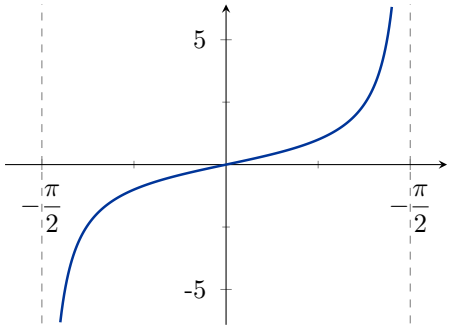

$y = \cos(x)$



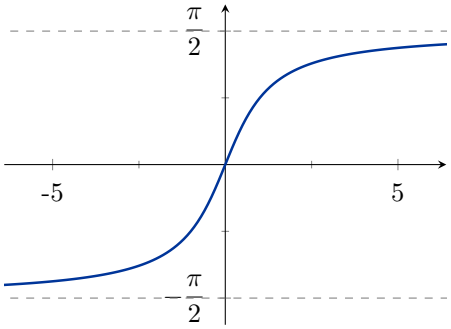

$y = \cos^{-1}(x)$




$y = \tan(x)$




$y = \tan^{-1}(x)$



**Example.** Evaluate the following:

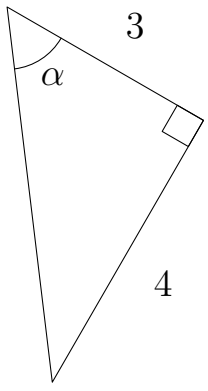
$\sin^{-1}\left(\dfrac{\sqrt{2}}{2}\right)$

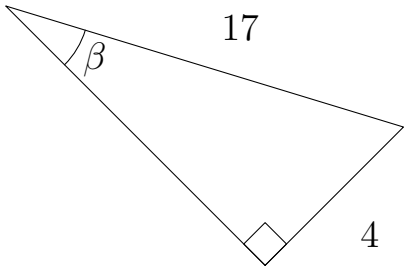
$\cos^{-1}\left(-\frac{1}{2}\right)$

$\tan^{-1}(-1)$

$\tan^{-1}\left(\dfrac{\sqrt{3}}{3}\right)$

**Example.** Use inverse trigonometric functions to solve for the following angles:



**Example.** Evaluate the following:

$\sin^{-1}\left(\sin\left(37^\circ\right)\right)$

$\cos^{-1}\left(\cos\left(\frac{9\pi}{4}\right)\right)$

$\sin^{-1}\left(\cos\left(\frac{40}{41}\right)\right)$

$\tan^{-1}\left(\sin\left(\frac{12}{20}\right)\right)$

**Example.** Simplify the following:

$\cos\left(\sin^{-1}\left(\frac{x}{3}\right)\right)$, where $-3 \leq x \leq 3$

$\sin\left(\tan^{-1}\left(4x\right)\right)$